Preference Versus Actual Choice: Evidence from a Payments Survey

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Use of payment methods at the POS

Four (4) groups of factors that influence use of payment methods:

Consumer-specific

1. *Preferred payment method.
2. Ownership of credit and debit cards.
3. Demographics.

Merchant-specific

1. Acceptance of payment methods.
2. Cash discount and card surcharge (**steering).

Transaction-specific

1. Payment $ amount.
2. Product/service type.

Bank/card-specific

Credit card rewards (**steering).

*Main topic of today's talk.

**Related topic.

The main research question

"You can't always get what you want." The Rolling Stones, 1969

In the Survey and Diary of Consumer Payments Choice consumers are asked:
What is your preferred payment method for nonbill in-person payments?

Research questions:

1. Do consumers make choices based on their stated preferences?
2. Can the stated preferred method predict actual use of a payment method?

Methodology: Apply machine-learning techniques to measure and quantify the degree of predictability of the stated preferred payment method.

Related literature: Steering payment choice

Merchants' attempts to steer consumers to pay with a payment method that is less costly to merchants:

- Bolt, Jonker, & Van Renselaar (2010), *J. of Banking and Finance*.
- Hayashi (2012), FRB–KC.
- Briglevics, & Shy (2014), *Review of Industrial Organization*.
- Welte (2014), Ph.D. dissertation, Carleton University.
- Stavins and Shy (2015), *J. of Behavioral & Experimental Economics*.
- Welte (2016), *J. of Financial Market Infrastructures*.
- Stavins (2018), *J. of Banking & Finance*.
- Greene, Shy, & Stavins (2026), FRB-ATL (to be discussed).

Note: Steering at the merchant level!

Related literature: Credit card reward

The effects of credit card rewards (such as cashback and travel miles):

- Ching & Hayashi (2010), *J. of Banking & Finance*.
- Carbó-Valverde, & Liñares-Zegarra (2011), *J. Banking & Finance*.
- Simon, Smith, & West (2010), *J. of Banking & Finance*.
- Arango, Huynh, & Sabetti (2015), *J. of Banking & Finance*.
- Agarwal, Presbitero, Silva, & Wix (2023), Board of Governors.
- Felt, Hayashi, Stavins, & Welte (2023). *J. of Banking & Finance*.

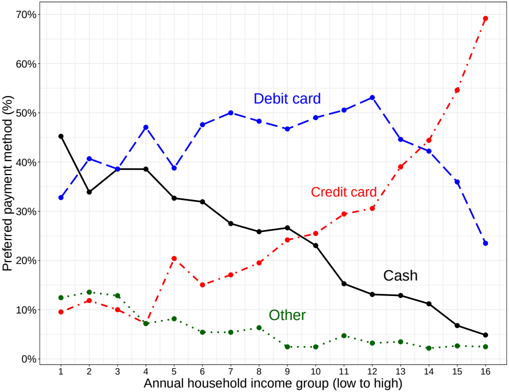
Note: Steering by financial institutions
(banks, card networks, but not merchants)!

Data

- 2025 Survey and Diary of Consumer Payment Choice (SDCPC).
- 6,079 respondents (individuals) who recorded 34,245 transactions.
- Some respondents or transactions were removed from the sample because of missing information.
- The sample was reduced to include only nonbill in-person payments.
- The analysis of actual payment choice used a subsample containing 17,177 nonbill in-person payments made by 4,747 survey respondents corresponding to 10 spending/merchant categories (out of 21 categories).
- The R-code and data are posted on my website (link to GitHub).

Consumers' preferred payment method

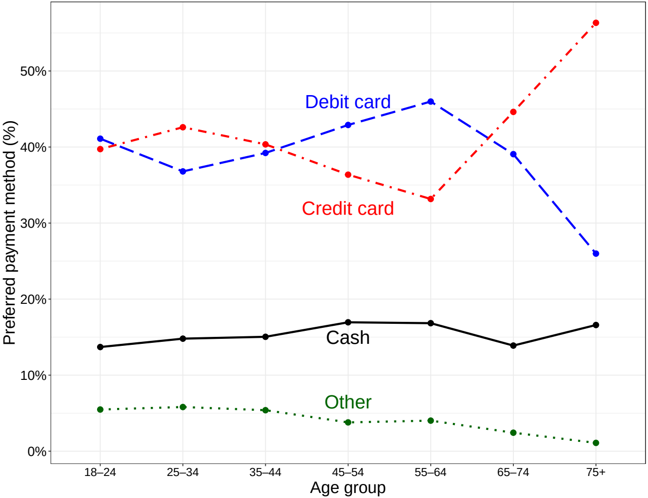
Preference by household income group (nonbill in-person payments)



1) Less than \$5,000; 2) \$5,000–\$7,499; 3) \$7,500–\$9,999; 4) \$10,000–\$12,499; 5) \$12,500–\$14,999; 6) \$15,000–\$19,999; 7) \$20,000–\$24,999; 8) \$25,000–\$29,999; 9) \$30,000–\$34,999; 10) \$35,000–\$39,999; 11) \$40,000–\$49,999; 12) \$50,000–\$59,999; 13) \$60,000–\$74,999; 14) \$75,000–\$99,999; 15) \$100,000–\$149,999; 16) \$150,000 or higher.

Consumers' preferred payment method

Preference by age group (nonbill in-person payments)



Consumers' preferred payment method

Credit card preference versus card ownership

Card ownership	Credit card Prefer	Credit card Not prefer	Debit card Prefer	Debit card Not prefer
Total	2411	3574	2387	3598
Own card	2396	2473	2349	3038
Do not own card	15	1101	38	560
Own card with reward	2340	1964		
Own card without reward	56	509		

- Credit card ownership $\not\Longrightarrow$ preference for credit card payments.
- Credit card ownership $\Longleftarrow$ preference for credit card payments.
- (applies also to debit cards).
- Owning reward credit cards $\Longrightarrow$ 20% higher preference for credit card.

Prediction accuracy: Confusion table

Preferred payment method (predictor)	Actually-used payment method: Cash	Credit	Debit	Other
Cash	1081	291	465	141
Credit	1168	5772	437	567
Debit	1193	614	4304	447
Other	173	186	219	119
Cash	55%	15%	24%	7%
Credit	15%	73%	6%	7%
Debit	18%	9%	66%	7%
Other	25%	27%	31%	17%
True Positives (TP)	1081	5772	4304	119
False Negatives (FN)	2534	1091	1121	1155
False Positives (FP)	897	2172	2254	578
True Negatives (TN)	12665	8142	9498	15325
Sensitivity (SEN)	0.30	0.84	0.79	0.09
Specificity (SPE)	0.93	0.79	0.81	0.96
Pos Pred Value (PPV)	0.55	0.73	0.66	0.17
Neg Pred Value (NPV)	0.83	0.88	0.89	0.93
Balanced Accuracy (BA)	0.62	0.82	0.80	0.53

TP= the number of payments made by respondents who actually used their preferred method.

$\text{PPV} = \frac{\text{TP}}{\text{TP+FP}}$

FP= see next slide.

Note: TP, FP, TN, and FN are computed for each payment method separately.

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FP (cash) = the number of non-cash payments made by respondents who prefer cash.

Example: FP (cash) $= 291 + 465 + 141 = 897$.

FN (cash) = number of cash payments made by respondents who prefer non-cash.

Example: FN (cash) $= 1168 + 1193 + 173 = 2534$.

Prediction: Classification trees

Consider the following model:

$$\overbrace{\text{Payment method}_{i,t}}^{\text{actually used}} \sim \overbrace{\text{Preference}_i + \text{Own credit}_i + \text{Credit reward}_i + \text{Own debit}_i}^{\text{individual specific}}$$

$$+ \underbrace{\text{Cash accepted}_{i,t} + \text{Credit card surcharge}_{i,t} + \text{Discount for payment method}_{i,t}}_{\text{transaction specific}}$$

$$+ \underbrace{\text{Amount (\$)}_{i,t} + \text{Merchant}_{i,t} + \text{Card accepted}_{i,t}}_{\text{transaction specific}}.$$

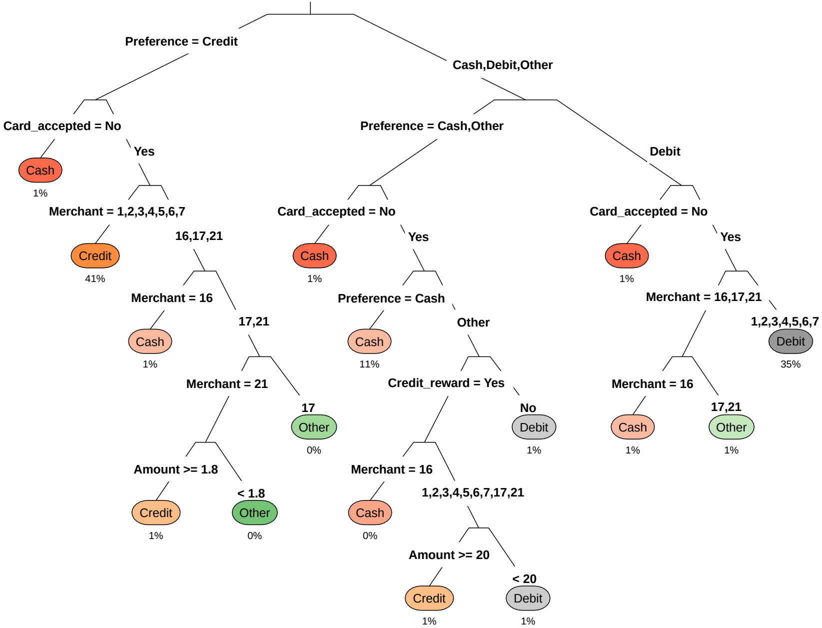
t denotes a transaction (payment) and i is an individual (respondent).

Payment method and preference can take 4 possible values: Cash, Credit card, Debit card, and Other.

The tree illustrates how the machine (software) splits and classifies the data according to the objective of minimizing (a function of) the number of payment method classification errors (impurity measures).

Conclusion: Preference for a payment method is the key predictor for actual use. Card acceptance ranks second, merchant-type third.

Prediction: Classification trees



Prediction: Random forest

- Random forest constructs a large number of classification trees.
- Each tree splits the data according to a randomly-selected subgroup of predictors.
- The number of predictors in each subgroup is controlled by the researcher.
- The number of excluded predictors is generally chosen to be near the square root of the total number of predictors, i.e. $\lfloor\sqrt{10}\rfloor = 3$.
- $\Longrightarrow$ random forest removes randomly 3 predictors from each tree.
- This procedure decorrelates the sampled trees so that any strong predictor (such as the preferred payment method) is not chosen as the top split in each tree.
- Trees cannot be drawn (hundreds of them).
- But, this algorithm makes (very good!) predictions.

Random forest: variable importance table

(a.k.a. variable importance plot, VIP)

	MDA for payment method used				Overall
Predicting variable	Cash	Credit	Debit	Other	MDA
Preferred payment method	64.11[2]	135.18[1]	146.68[1]	-1.64	150.76[1]
Spending/merchant category	50.55[4]	67.25[3]	63.47[3]	62.18	96.95[2]
Spending amount	38.42	28.53	26.51	36.13	48.20
Own credit card	9.41	13.94	-6.90	13.96	16.68
Credit card reward	6.74	26.53	8.02	10.41	29.44
Own debit card	0.70	-8.72	49.52[4]	6.53	48.76
Card accepted	103.77[1]	109.35[2]	95.15[2]	27.31	136.47[2]
Cash accepted	54.61[3]	6.43	8.50	6.79	48.01
Credit card surcharge	23.97	41.95[4]	46.91	9.83	52.63[4]
Discount for payment method	27.81	29.68	36.71	15.94	48.71

MDA (Mean Decrease Accuracy) measures by how much prediction accuracy is reduced with the elimination (permutation) of each of the ten predictors separately. Hence, the degree of variable importance.

Random forest: Prediction accuracy

(Comparing random forest predictions to raw data predictions)

Preferred payment method (predictor)	Actually-used payment method Cash	Credit	Debit	Other
Cash	504	63	115	67
Credit	237	1784	162	139
Debit	298	193	1360	137
Other	19	10	10	56
Cash	67%>55%	8%	15%	9%
Credit	10%	77%>73%	7%	6%
Debit	15%	10%	68%>66%	7%
Other	20%	11%	11%	59%>17%
True Positives (TP)	504	1784	1360	56
False Negatives (FN)	554	266	287	343
False Positives (FP)	245	538	628	39
True Negatives (TN)	3851	2566	2879	4716
Sensitivity (SEN)	0.48	0.87	0.83	0.14
Specificity (SPE)	0.94	0.83	0.82	0.99
Pos Pred Value (PPV)	0.67>0.55	0.77>0.73	0.68>0.66	0.59>0.17
Neg Pred Value (NPV)	0.87	0.91	0.91	0.93
Balanced Accuracy (BA)	0.71>0.62	0.85>0.82	0.82>0.80	0.57>0.53

Conclusions

- Stated preferences for credit and debit cards are the key predictors of credit and debit card payments.
- Stated preferences for paying cash is the second more important predictor of actual cash payments.
- For cash, card acceptance is the key predictor.
- Random forest predictions improve accuracy especially for "cash" and "other" payments.

Note on random forest: Predictions are made on a "testing" subsample which is separate from the "training subsample."